J. Earth Syst. Sci. (2024)133 186
https://doi.org/10.1007/s12040-024-02378-0

© Indian Academy of Sciences

Integration of machine learning and remote sensing for assessing the change detection of mangrove forests along the Mumbai coast

Suraj Sawant[1,*], Praneetha Bonala[1], Amit Joshi[1], Mahesh Shindikar[2], Abhilasha Patil[3], Swapnil Vyas[3] and Deepti Deobagkar[4]

[1]*Department of Computer Science and Engineering, College of Engineering Pune Technological University, Wellesley Road, Shivajinagar, Pune 411 005, India.*
[2]*Department of Applied Sciences and Humanity, College of Engineering Pune Technological University, Wellesley Road, Shivajinagar, Pune 411 005, India.*
[3]*Department of Geography (Geoinformatics), Savitribai Phule Pune University, Pune 411 007, India.*
[4]*Formerly: Department of Geography (Geoinformatics), Savitribai Phule Pune University, Pune 411 007, India.*
**Corresponding author. e-mail: sts.comp@coeptech.ac.in*

MS received 26 December 2023; revised 16 April 2024; accepted 19 April 2024

Mangrove forests, being high-yielding ecosystems, often dominate the intertidal sites along equatorial and subtropical coasts. Despite the known significance of mangroves to the coastal ecology, especially fisheries, deforestation remains a severe danger due to coercion for forest products, ground transformation for aquaculture, and seaside urban growth. Remote sensing is integral in mapping and analysing changes in mangrove forests' areal extent and spatial patterns due to natural disasters and anthropogenic causes over the last three decades. This work depicts remote sensing analysis for change detection in mangrove forest land use land cover from 2014 to 2019. Indian Remote-Sensing Satellite Resourcesat-2 LISS-IV datasets have been used for analysis. A comparison with the Sentinel-2A dataset and two machine learning models: Random Forest and Classification and Regression Tree, has been performed with 2019 data. This work identifies CART as a suitable choice for supervised landform classification utilising remotely sensed geophysical data that is used to decipher spatial changes concurred over time. An overall growth in the mangrove cover was observed from 2014 to 2019, from 86.26 to 89.63 km^2, along the Mumbai coastline. Spatial comparison over the years shows the growth and loss of land-use cover areas. The performance metrics such as overall accuracy, producer accuracy, Kappa coefficient, and Matthews correlation coefficient are computed. The experiments were conducted using the Google Earth Engine, a powerful cloud computing platform.

Keywords. Mangrove forest; Machine Learning Algorithms; land use land cover; assessment; change detection; Google Earth Engine.

1. Introduction

Mangrove forests that are found geographically along seashores have evolved to become the most important component of the coastal ecosystem (De Silva and Amarasinghe 2023). They have numerous advantages in terms of maintaining the environment's ecological equilibrium. They act as buffers between land and sea, protecting the land from the effects of ocean hazards such as floods and

Published online: 16 September 2024

preventing soil erosion by capturing and settling tide-borne sediments (Carlson *et al.* 2021). They are environmentally friendly and safeguard coastal areas against harmful solar radiation, floods, greenhouse gases, storms, sea-level rise, and coastal soil erosion. Mangroves benefit many aquatic species by providing feeding, breeding, and nursery grounds (Arceo-Carranza *et al.* 2021). These mangroves provide significant ecological benefits to human society as well as coastal and marine environments (Silva *et al.* 2014; Bimrah *et al.* 2022). They are economically important as a source of forestry and fishing products that help in the growth of the eco-tourism industry (Astikasari *et al.* 2023). Mangroves have an increased bioprospecting prospect as a source of salt-tolerant genes, compounds, and useful by-products for use in the pharmaceutical, industrial, agricultural, and food industries (Budiyanto *et al.* 2022; Nag *et al.* 2024). Regardless of their diverse value, these mangrove forests have suffered a global loss of more than 35%. According to the reports shared by the Forest Survey of India (https://www.fsi.nic.in/forest-report-2019) for years 2014–2019, there has been a significant increase in the mangrove forests along the Maharashtra coastline, particularly along the Mumbai city shoreline.

Remote Sensing (RS), together with Geographic Information System (GIS), a geospatial analyzing tool such as ArcGIS and QGIS (Sikakwe 2023; Singh *et al.* 2024) has major importance. It has significant applications in understanding various environmental aspects like waste management (Singh 2019), soil-erosion (Phan and Stive 2022; Nguyen *et al.* 2023), water management (Sulochanan *et al.* 2022), agriculture (Padhy *et al.* 2022), crop management (Kadaverugu *et al.* 2022; Trialfhianty *et al.* 2022), land-use classification (Faruque *et al.* 2022) and many others. Various satellite imagery acquired from satellites such as Landsat-7/8, Sentinel-2A, IKONOS, LIDAR, LISS-IV, and aerial images, using drones, have been used to perceive the land/sea patterns of the earth's surface. These satellite images can be acquired from any geological web portals such as United States Geological Survey (USGS) (https://www.usgs.gov/), European Space Agency (ESA) (https://www.esa.int/), Bhuvan (https://bhuvan.nrsc.gov.in/home/index.php), Bhoonidi (https://bhoonidhi.nrsc.gov.in/bhoonidhi/home.html) and Google Earth Engine (https://earthengine.google.com/), a cloud computing platform. The progressions that have happened over a period can be examined and observed utilizing the current and past satellite information (Wulder *et al.* 2022; Sameer and Hamid 2023).

Machine learning is another popular method for comprehending land-cover metrics. For mangrove conservation and mapping, various machine learning techniques (Quang *et al.* 2022) have been considered for feature extraction (Zhao *et al.* 2023), mapping of land-cover classes (Wang *et al.* 2022), and detecting changes over time (Goswami *et al.* 2022). With the advancement of technology, few state-of-the-art traditional supervised learning algorithms like Maximum Likelihood Classification (MLC) (Soni *et al.* 2022; Alshari *et al.* 2023) and Principal Component Analysis (PCA) (Jardim *et al.* 2022; Toosi *et al.* 2022; Xu and Deng 2022), and unsupervised classification technique Iso Cluster algorithm (Evan *et al.* 2023; Islam *et al.* 2024; Li *et al.* 2024), have shown significant results in computing the change analysis and other performance metrics measures.

Many advanced machine learning algorithms such as Random Forest classifier (Purwanto *et al.* 2022; Miraki *et al.* 2023; Zhao *et al.* 2023), Support Vector Machine (SVM) (Nagarajan *et al.* 2022; Priyono *et al.* 2022; Purwanto *et al.* 2022; Yang *et al.* 2022) are widely used with remote sensing. Machine learning has proven beneficial for a large number of applications in numerous domains of the globe and beyond, from estimating soil moisture (Xia *et al.* 2022), flood and landslide susceptibility assessment (Ado *et al.* 2022; Saravanan *et al.* 2023), and crop yield (Morales and Villalobos 2023).

CART algorithms generally determine complex interactions among variables in the final tree. CART works by recursively separating data until predetermined terminating points or terminal nodes are reached. CART begins by examining all descriptive variables and finding which binary division of a single description variable minimizes response variable deviation the most (Lawrence and Wright 2001). On the other hand, the use of Random Forest is instigated as a predictor or classifier and decided by taking a plurality vote among all the predictors. It is an ensemble method of multiple CART-like classifiers. It has been demonstrated in bagging that as the number of predictors increases, accuracy increases until it reaches a threshold when it drops off. The highest accuracy will be achieved by determining the optimal number of predictors to create. By bagging remotely sensed data and employing several decision trees, they were able to improve classification

accuracy (Kulkarni and Lowe 2016). Based on the literature survey, the use of these two algorithms has been used profusely for mangrove classification. In line with the literature review, CART and RF have been applied to the proposed methodology of this work.

The primary goal of the current work is to explore the changes in the mangrove cover along the west coast of Maharashtra coastline, along the city of Mumbai. Specifically, this paper seeks to (1) compare the classification accuracies of LULC maps produced from LISS-IV and Sentinel-2A data collected for the year 2019, (2) provide spatiotemporal change analysis of mangrove cover over five years, from 2014 to 2019, using LISS-IV data, and (3) investigate the most critical socio-economic drivers and predictors of growth and decline in mangrove cover. Confusion matrix, overall accuracy (OA), producers accuracy (PA), Kappa coefficient (K), Matthew's correlation constant (MCC), and an area of each land-use class are used to measure each of these objectives.

The design of this paper is as follows: section 2 describes the study area, image preprocessing, classification steps; section 3 deals with the methodology and defining accuracy metrics and validation. Section 4 provides results of the objectives described above. It shows the implementation of Random Forest and CART for classifying pixels for multispectral images, i.e., LISS-IV and Sentinel-2A. A comparison of their performances is made, reflecting the choice of the CART algorithm for change detection. This is further implemented on LISS-IV data for the years 2014 and 2019. Subsection 4.4 and section 5 provide a discussion of the findings and conclusion.

2. Dataset

2.1 *Study area*

Mumbai, India's largest metropolis, is one of the most populous cities in the country, found along the coast of Maharashtra, India. This region is geographically located at latitude 19.0760°N and longitude 72.8777°E covering about 521.26 km^2 surface area. According to the current research, mangroves along this shoreline have had a substantial impact on the infrastructure. Many mangrove creeks are found along its shoreline, and for this study, we will focus on the following creeks: Dharamtar Creek, Thane Creek, and Panvel Creek, as shown in figure 1.

2.2 *Image preprocessing*

Two cloud-free scenes of IRS Resourcesat-2 LISS-IV satellite data (1st week of March for years 2014 and 2019) are acquired for the Mumbai region from the National Remote Sensing Centre (NRSC), Bhoonidhi web portal. These images are geometrically and atmospherically corrected. This sensor data has a high-resolution multi-spectral sensor operating in three spectral bands: Green (0.52–0.59 mm), Red (0.62–0.68 mm), and Near-Infrared (0.77–0.86 mm) and provides a resolution of 5.8 m. Sentinel-2A satellite data, for the same time frame, are collected from Google Earth Engine (GEE), which provides a resolution of 10 m as shown in table 1. The land-use land cover maps are computed for this data using GEE (Shetty 2019) web-based JavaScript API.

Cloud cover for these images was handled at 0%, to avoid any discrepancy in the land use classification results. Classification of these collected images is performed on Google Earth Engine as explained in section 2.3.

2.3 *Image classification*

This work identifies four natural land-use classes: (1) mangroves, (2) water, (3) wetlands, (4) other (includes buildings, roads, and bareland), well suited for the study area. Table 2 shows the training samples collected for each class through separate equalised random segmenting in the form

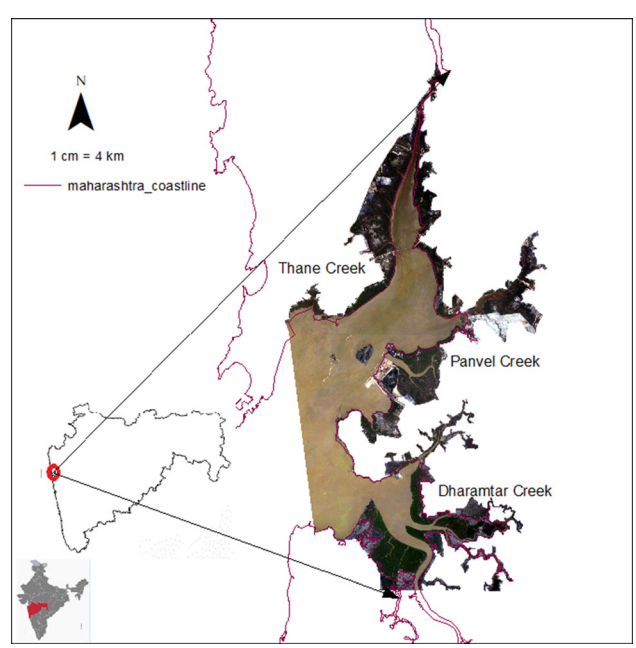

Figure 1. Study area.

Table 1. *Information of satellite data used.*

Satellite	Date	Spectral bands considered	Spatial resolution (m)	Bands
LISS-IV	First week of March for year 2014	3	5.8	Green, Red, Near-infrared
Sentinel-2A	First week of March for year 2019	4	10	Blue, Green, Red, Near-infrared

Table 2. *Training samples.*

Study area	Land-use class	No. of polygons	Total training samples
Mumbai region	Mangroves	35	150064
	Water	36	
	Coastal wetlands	26	
	Other	24	

of polygons. Every possible care is taken to have a uniform distribution of the training dataset to make balanced samples of classes. The categories of the concerned pixels are specified by visually analysing the imagery using high-resolution images from Google Earth Pro. These spectral signatures or training samples, are collected from the True Composite Image (layered band image of bands Red, Green, and Blue) of Sentinel-2A as shown in figure 1. The polygons collected for each land-use type depend upon the exposure the study area portrays. Mangrove and water classes had more visibility; therefore, 35 and 36 polygons were collected, respectively, whereas wetlands and other classes had less visibility. Therefore, 26 and 24 polygons were collected.

3. Methodology

The overall workflow of the research work is represented in figure 2. The methodology identified for this work is finalised based on the literature review done to minimise the complexity of the computations and to get enhanced results. For this land-use categorisation, the most extensively used decision tree technique, known as CART, is used as the machine learning strategy. It shows the capability to acquire rules from training samples, needed for classification. Thus, retaining the benefit of being self-dependent on the hypotheses of value distribution, i.e., can access variables independently from each other (Breiman *et al.* 1984). Pixels are divided into two parts, at 70% and 30%, to clear any possible inclinations driven by them to evaluate the classifiers.

LISS-IV data is collected for years 2014 and 2019, keeping cloud cover equivalent to 0%. The temporal data considered is of the first week of March for both the study years to be able to provide proper change analysis for five years.

3.1 *Algorithms*

A well-known classifier in machine learning is the Classification and Regression Trees (CART) algorithm, which is mostly used for decision tree learning. Charles Stone *et al.* first introduced it in the 1984 book 'Classification and Regression Trees' (Leo *et al.* 1984). Because of its ease of use, interpretability, and efficiency in both classification and regression tasks, CART is extremely popular. CART uses recursion to construct a binary tree. It determines the feature and split point that best divides the data into two subgroups, maximising the homogeneity within each subset, starting at the root node. This procedure is carried out recursively for every subset until a halting condition is satisfied. CART assesses the quality of splits using impurity metrics like mean squared error for regression tasks and Gini impurity for classification tasks. At each stage, it chooses the split that minimises mistakes or impurity. The tree-building process ends when one of the predetermined halting conditions is satisfied. A maximum tree depth, fewer samples than a minimal threshold at a node, or the point at which additional splitting does not appreciably increase prediction accuracy are standard stopping criteria. Pruning procedures can be used after the tree is built to stop overfitting. Removing branches from the tree that do not

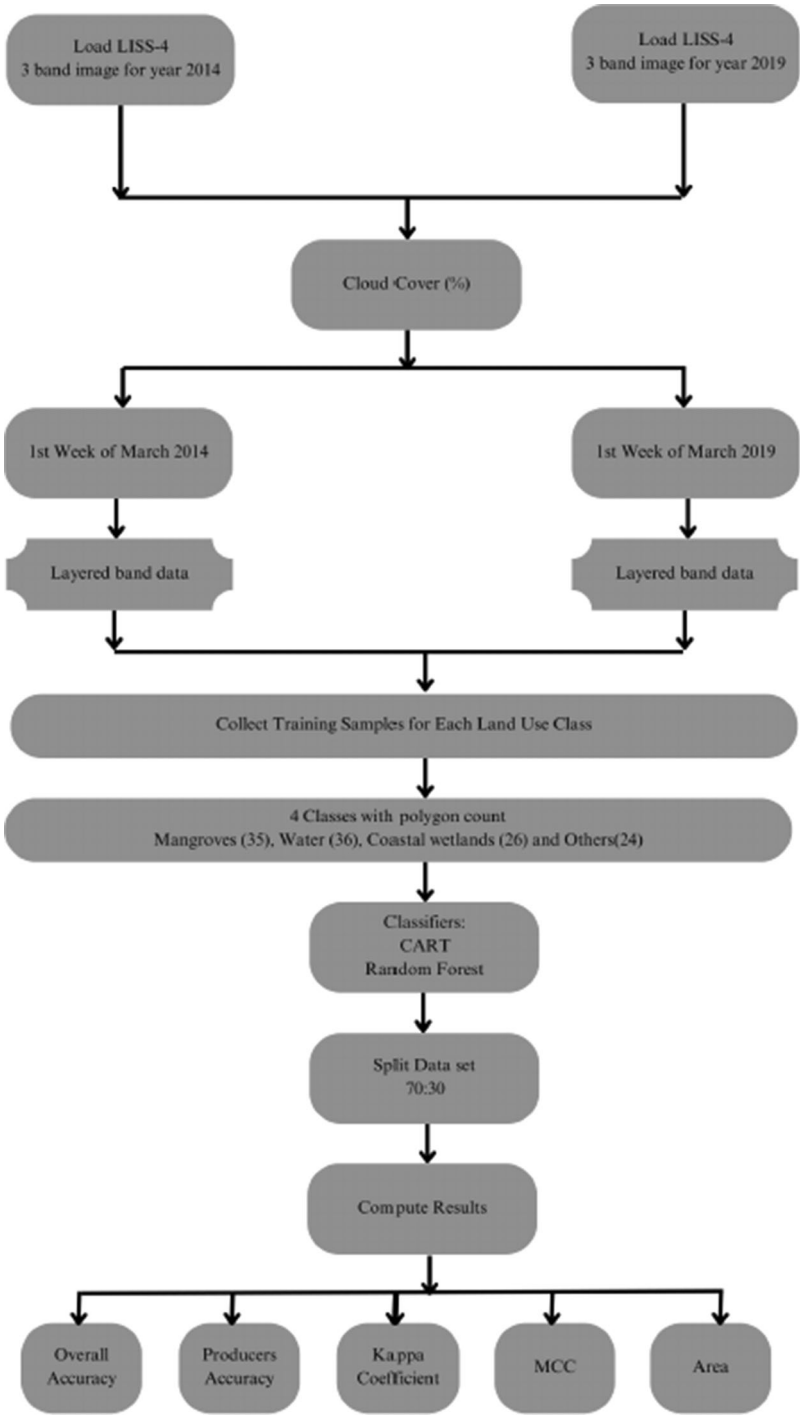

Figure 2. Work flow.

considerably increase prediction accuracy on a validation dataset is known as pruning. After the tree is built, it may forecast new data by going through the tree from the root node to the leaf node corresponding to the input data's feature values. Since CART is a non-parametric technique, it does not make any assumption on the data's underlying distribution. Because of its adaptability, CART can capture intricate connections between target variables and input data without making arbitrary assumptions (Al-Awar *et al.* 2022).

A prominent ensemble learning technique that expands on decision trees' basis is known as Random Forest. Adele Cutler and Leo Breiman (Breiman 2001) presented it for the first time in 2001. Random Forest generates multiple random subsets of the original dataset using a process known as bootstrapping. Every subset, referred to as a

bootstrap sample, is produced by sampling and replacing the original data. As a consequence, many training datasets are produced that are identical in size to the original dataset but contain duplicate and missing observations. Random Forest chooses a random subset of features from the whole feature set at each decision tree node. The selection of a random subset reduces the connection between the trees in the forest and helps to introduce variety among them. One hyper-parameter that may be adjusted is the number of features taken into account at each split. A decision tree is trained with a subset of the features and a portion of the data for every bootstrap sample. The recursive partitioning method used to create standard decision trees, such as CART, is also used to grow this decision tree. The decision tree is usually pruned only up to a certain depth to avoid overfitting. After every tree has reached maturity, forecasts are produced by adding up each tree's forecasts. For classification tasks, the class predicted by the majority of trees (mode) is assigned to the input data point. In regression problems, the final prediction is the mean of all the trees' predictions. Random Forest is well known for many machine learning tasks because of its high accuracy, resilience, feature selection capabilities, and scalability (Demir and Şahin 2022). This is especially true in real-world applications where prediction performance and model interpretability are crucial.

confusion matrix is used to calculate OA (Congalton 1991; Wang *et al.* 2020), PA (Congalton 1991; Akca and Polat 2022), K (Wieckowska *et al.* 2022) and MCC (Chicco and Jurman 2023a, b). The area of every land-use class is computed to provide an analysis on the change cover.

High-resolution images from Google Earth are commonly utilised for ground truth data that is used to assess categorisation precision (Estoque *et al.* 2018). Due to the tribulation in getting ground values of mangrove forest distribution over a broad scale, images emanated from Google Earth are used as authentic references to evaluate the OA, PA, K, and MCC. By referencing to the locations of the mangrove areas, the existence of them on high spatial-resolution images is validated. Equations (1–4) show the mathematical representation of the accuracy measures considered.

$$OA = \frac{TP + TN}{TP + FP + TN + FN} \tag{1}$$

$$rPA = \frac{TP}{TP + FN} \tag{2}$$

$$K = \frac{TA - p_e}{1 - Pe_e},$$

where

$$Pe_e = \frac{(TP + FN) * (TP + FP) + (FP + TN) * (FN + TN)}{(TP + FN + TF + FP)^2} \tag{3}$$

$$MCC = \frac{TP \times TN - FP \times FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \tag{4}$$

3.2 *Evaluation metrics*

After classification, the correctness of these classification results was confirmed. In this study, accuracy measurements are utilized to assess the quality of the classification. These metrics compare the classification result to on-site ground values, that are dependable and accurate. Evaluating a confusion matrix provides all related relations between the classification and referenced data. The

For the first objective, OA, K, and the area of mangrove cover of the multivariate classifications of the year 2019 are calculated and compared for Sentinel-2A and LISS-IV, respectively. Their performance is analysed using Random Forest and CART. The resultant model is further used to achieve the second objective of change detection. OA, PA, K, MCC, and area of every land-use class are computed and analysed.

Table 3. *Comparative analysis of mangrove cover for Sentinel-2A and LISS-IV images.*

	Random Forest			CART		
Type of satellite	Area (km^2)	OA	K	Area (km^2)	OA	K
Sentinel-2A	81.65	99.98	0.99	82.22	100	1
LISS-IV	104.75	99.55	0.98	89.63	99.66	0.98

Although OA and PA cannot guarantee the performance of each class, K and MCC results are used as decision-makers providing adequate analysis in terms of acceptable and agreeable classification of the land-use classes for the proposed model. The kappa (K) value is a proportion of how closely the classes ordered by the machine learning classifier match the information known as ground truth while controlling the precision of a random classifier as estimated by the normal exactness. It ranges between 0 and 1, with 1 denoting perfect acceptance and a value near zero, implying that the acceptance is nothing more than a coincidence. MCC values are typically between −1 and +1, indicating the rate of agreement between the classified points and the training samples (Yao and Shepperd 2020). This metric provides a better description of the results obtained after classification than F1-score and accuracy, so it is also used for evaluation (Chicco and Jurman 2020).

4. Results and discussions

The classifications of mangroves and other land use classes emanated from satellite images using machine learning algorithms. Based on the proposed objectives, results are computed, analysed, and concluded.

4.1 *Experimental setup*

The whole classification strategy is developed within the GEE environment using web-based JavaScript API. The performance computed here is given in terms of Confusion matrix, OA, PA, K, and area of each land-use class. MCC is computed within Jupyter Notebook using Python implementation. The code will be made available as per request to the corresponding author.

Every training sample was collected from the GEE *Map* region. Using *Polygon Geometry tool*, spatial information of each land-use class was collected as a *Feature Collection*. LISS-IV study regions of years 2014 and 2019, which are extracted using ArcGIS 10.3, are uploaded to *GEE Assets*, as GeoTIFF files.

4.2 *Analysis of Sentinel-2A and LISS-IV results*

The intercomparisons of the performance metrics inferred from both Sentinel-2A and LISS-IV images are carried out. Table 3 shows the results of the classification for the year 2019, using Sentinel-2A data of 10 m resolution and LISS-IV data of 5.8 m resolution for the categorical mapping. An accuracy assessment is performed using 150064 pixels, for all classes combined. These imagery data are trained using the two most popular machine learning models for LULC classification: RF and CART.

The model accuracy for Sentinel-2A satellite data using RF and CART showed 99.98% and 100%, respectively, whereas LISS-IV computed its accuracies at 99.55% and 99.66%, respectively. The K is found to be between 0.98 and 1 for all the cases. Using Sentinel-2A, the mangrove area calculated is found to be 81.65 and 82.22 km^2, respectively. With the same spectral signatures, LISS-IV computed the mangrove cover as 104.75 and 89.63 km^2, respectively. From these results, it can be inferred that area calculation of the mangrove cover using LISS-IV imagery, executed along with the CART algorithm, provides a value closer to that of results obtained using Sentinel-2A. Thus CART algorithm extrapolates the importance of its use for LULC using LISS-IV satellite images.

Although any satellite image could be used for land-use classification, the one with high image resolution, i.e., LISS-IV, showed the best possible results. Figure 3 shows the spatial differences perceived in the classification of both satellite imagery using the CART algorithm.

There are only a few visible differences between the classified LISS-IV and Sentinel-2A images. LISS-IV detected the wetlands in the region more precisely than Sentinel-2A. Another interesting discrepancy observed at Elephenta Island is the position of mangroves on an elevation in

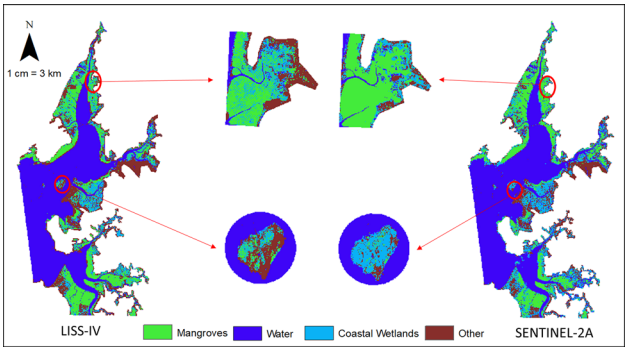

Figure 3. LISS-IV and Sentinel-2A results comparison.

Sentinel-2A, which is terrestrial vegetation, but mangroves are only identified at the shorelines in LISS-IV classified data. In this region, hilly areas are also classified as wetlands in Sentinel-2A (as shown in figure 3), whereas the same land area is correctly classified as Other in LISS-IV.

Anticipating the differences in the image resolution of LISS-IV and Sentinel-2A, this pixel-wise classification, performed using CART algorithm reads the training samples and assigns them to different land-use classes based on the pixel readings of each band. These discernible differences show that the results obtained with LISS-IV are more accurate, not only in terms of visual interpretation but also of image resolution and model used.

4.3 *Change detection*

With results and analysis provided and proved in section 4.2, CART algorithm shows better land use classification results. This model is further considered for the change detection. Further considering both spatial and temporal changes for the years 2014 and 2019, a change analysis is performed. Accuracies of the classification performed on the above-mentioned land use classes are shown in table 4.

The PA for the mangrove class showed 99.6% for the year 2014, whereas it showed 99.63% for the year 2019. Thus, values obtained for mangrove forests are attained with admissible accuracy. Accuracy results of the remaining classes are depicted in table 4, also showing justified accuracies. The OA is greater than 99%, and K of the acquired classified results are between 0.98 and 1, denoting ideal to immaculate its concurrence. The MCC values of the results are between 0.89 to 1, indicating perfectly agreeable classified results. Differences along the five years are shown graphically in figure 4 for better understanding.

The spatial pattern of the land-use classes over five years is shown in figure 5. The classified results indicate that mangrove distribution displayed a similar spatial pattern in this tenure. Mangrove forests distributed along three creeks, of the Mumbai coast were considered as study areas for the present work. Thane Creek is seen in the north, Panvel Creek in the east, and Dharamtar Creek is found in the south part of the study area as illustrated in figure 1.

In figure 5, it could be seen (through visual interpretation) that in 2019, the mangroves cover in Thane Creek showed no significant increase in mangrove cover, but showed a significant decrease in Panvel Creek and an increase in Dharamtar Creek. A few of these observable spatial changes

Table 4. *Accuracy indices assessed from the classification image in 2014 and 2019.*

Performance measures	2014	2019
Overall accuracy	99.71	99.66
Kappa coefficient	0.9869	0.9847
Producers accuracy		
Mangrove	99.6	99.63
Water	99.9	99.99
Wetlands	86.92	83.39
Other	97.92	99.82
Matthew's correlation constant		
Mangrove	1.0	1.0
Water	1.0	1.0
Wetlands	0.91	0.89
Other	1.0	1.0

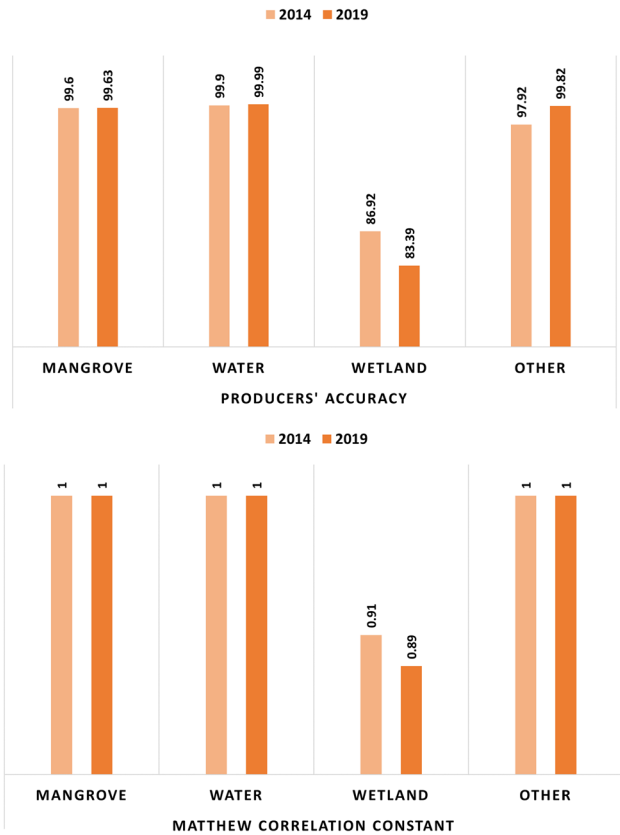

Figure 4. Performance metrics.

are highlighted in figure 5. The decline observed near Panvel Creek can be attributable to the expansion of the built-up region, which is part of the Other land-use class. In 2014, the Panvel Creek region showed a presence of wetlands and mangroves, which in 2019, shows the expansion of the Other (built-up regions) class. This results in the drifting of sediments that are getting settled in lower regions, offering a niche for the growth of mangroves. This is observed in the Dharamtar region. Also, the increase of the built-up region at Dharamtar Creek is attributed to the decline of wetlands in this region.

Further, analysis in view of the area covered by each land use class could be seen in table 5. Over the course of five years, land cover along the shoreline shows a notable increase in the mangrove cover by 3.37 km^2. Other class region, including both landward (e.g., urban and roads) and seaward (e.g., barren lands) sides, shows an abrupt increase in their area. From 72.54 km^2 in 2014 to 101.19 km^2 in 2019, which is probably due to the high proportion of built-up area growing towards the landward side of the region, near Panvel Creek. Wetlands, on the other hand, showed a significant decrease in its area, with 88.57 km^2 in 2014 to 68.12 km^2 in 2019, showing the effects of an increase in Other land use class, i.e., aptly showing the increase in barren land near ocean waters.

Differences are observed with necessary measures taken while performing ground analysis of our study region. This analysis is conducted using Google Earth Pro, which has premium high-resolution images, that help in finding the most intrinsic details of our study area with utmost sophistication. Images from Google Earth are commonly used to evaluate categorisation accuracy and might be considered for visual interpretations.

4.4 *Discussions*

The current study seeks to comprehend the spatio-temporal changes in mangroves that occurred in the Mumbai region in recent years (between 2014

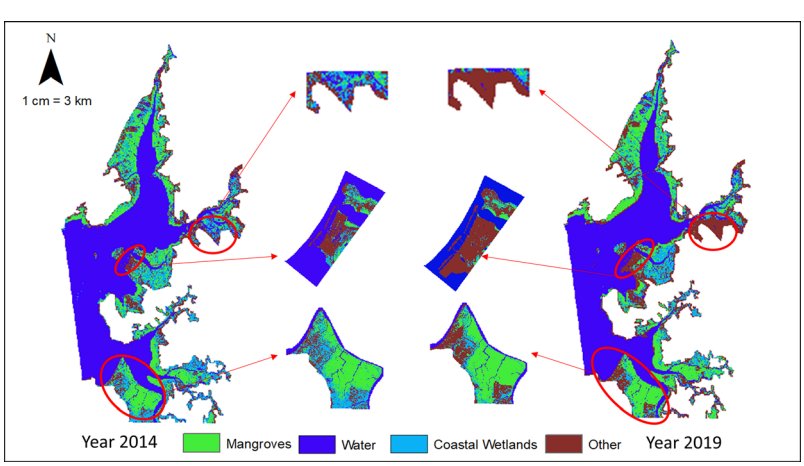

Figure 5. Change detection from year 2014 to 2019.

Table 5. *Land use/cover change for Mumbai region from 2014 to 2019.*

	2014		2019		
LULC class	Area (km^2)	Area (%)	Area (km^2)	Area (%)	Change in area from 2014 to 2019 (km^2)
Mangrove	86.26	0.165484	89.63	0.171945	3.37
Water	273.89	0.525438	262.33	0.503252	–11.56
Wetlands	88.57	0.169915	68.12	0.130681	–20.45
Other	72.54	0.139163	101.19	0.194122	28.65

and 2019). Images from the LISS-IV satellite, having a resolution of 5.8 m, are used for the change detection in spatial features. This area is classified into four land-use classes, using two machine learning models, Random Forest, and CART, covering the study area. These classified spatially distributed data are compared with results obtained from Sentinel-2A data. These results include the performance metrics such as OA, K, and area of mangrove cover. While Sentinel-2A calculated the mangrove cover of around 80 km^2, LISS-IV showed varying results for both the machine learning models. For the same training samples collected for computing the mangrove cover, LISS-IV showed 104.75 km^2 using Random Forest, whereas it showed 89.63 km^2 using CART. These results prompted the use of the CART algorithm as a more suitable machine learning model for further change analysis. The classification results of both LISS-IV and Sentinel-2A show that LISS-IV classifies wetlands better than Sentinel-2A. LISS-IV also differentiated mangroves from other vegetation features, which further proved its liability in choosing LISS-IV over Sentinel-2A for land-use classification.

LISS-IV satellite data for 2014 and 2019 were collected, having the same timestamp. This satellite data was classified, using the CART algorithm, into four land-use classes. According to the findings, the mangrove cover increased from 86.26 to 89.63 km^2. Other land-use types, primarily built-up and barelands, also showed an increase in their land-cover area. This increase is due to the increase in anthropogenic activities causing an increase in sediments, which are carried further down the sea and settle on the ocean shorelines, thus providing more land for mangroves to grow and thrive. The OA and K values are computed as 99.71% and 0.9869 for 2014 and 99.66% and 0.9847 for 2019. The PA for each class shows good results ranging between 83.39% and 99.8% for the land-use classes. The computed MCC values revealed that the classified samples recorded greater than 0.8 when all land-use types and study years were considered. This further proves the liability of our classified samples and model.

Compared with other mangrove vegetation areas in India, creeks along the Maharashtra coastline on India's West Coast have less diversity of mangroves. The ground surveys confirmed this. The zonation of mangrove forests in the study area is still far from defined. The overall mangrove vegetation is dominated by the species *Avicennia marina*, *Sonneratia apetala*, *Acanthus ilicifolius*, and *Excoercaria agallocha*, with sporadic occurrence of *Rhizophora mucronata*. *Avicennia marina* is very resilient to various stresses such as organic pollution, chemical effluents, and extreme salinity. In the southern part of our study region *Sonneratia apetala* is found to be more resilient in the upstream region of the creeks. This species also has a remarkable capacity for coexistence (Shindikar 2006). As a result, the entire mangrove forest exhibits a consistent spectral signature.

5. Conclusion

In this paper, the spatio-temporal analysis of the various land use classes along the Mumbai coastline is carried out. The research conducted is based on its distribution and variation pattern. This analysis is provided using Google Earth Engine (GEE). For two machine learning models, Random Forest and CART, the LISS-IV classified data for 2019 are compared to the Sentinel-2A classified data. Results show discernible changes in the classification, with coastal wetlands showing a notable presence in LISS-IV that Sentinel-2A missed. Compared to the ground-truth visual analysis performed with Google Earth Pro, the CART algorithm produced the best results. This algorithm is further used to classify data for 2014 and 2019. Over time, spatial changes showed an overall increase in mangrove cover, with an

expansion in built-up sites, exhibiting the concept of area increase. Hence contributes to detecting changes in mangrove cover and other land-use classes.

Thus, accurate and timely changes in LULC elements are critical for comprehending the relations and interactions between human and natural phenomena to make better decisions for their prosperity and sustainability. On the other hand, land-cover data provides critical information for sustainable natural resource management. Mangroves, the natural coast guards, provide immense ecological and economic value for the coastal sustenance and hence to be conserved on priority. As a result, the current study provides baseline data for mangrove forest management.

The future scope of this work is to apply Deep Learning architectures for classifying the mangrove area in order to get enhanced accuracy. A dataset of mangrove and non-mangrove images is to be generated so that different deep-learning architectures can be implemented and compared.

Author statement

SS: Conceptualization, data curation, formal analysis, visualisation, writing – original draft, review, and editing; PB: Formal analysis, data curation, software, visualisation, writing – original draft, review and editing; AJ: Supervision, writing – review and editing; MS: Formal analysis, writing – review and editing; AP: Data curation; SV: Data curation, writing – review and editing; and DD: Writing – review and editing.

References

Ado M, Amitab K, Maji A K, Jasińska E, Gono R, Leonowicz Z and Jasińska M 2022 Landslide susceptibility mapping using machine learning: A literature survey; *Remote Sens.* **14(13)** 3029.

Akca S and Polat N 2022 Semantic segmentation and quantification of trees in an orchard using UAV orthophoto; *Earth Sci. Inform.* **15(4)** 2265–2274.

Al-Awar B, Awad M, Jarlan L and Courault D 2022 Evaluation of nonparametric machine-learning algorithms for an optimal crop classification using big data reduction strategy; *Front. Artif. Intell.* **5(3)** 141–153.

Alshari E A, Abdulkareem M B and Gawali B W 2023 Classification of land use/land cover using artificial intelligence (ANN-RF); *Front. Artif. Intell.* **5(964)** 279.

Arceo-Carranza D, Chiappa-Carrara X, Chávez López R and Yáñez Arenas C 2021 Mangroves as feeding and breeding grounds; *Mangroves: Ecology, Biodiversity and Management,* pp. 63–95.

Astikasari L, Indriyani S, Muryanto B S, AL Madani A R, Muhammad F, Putri A, Hartanti A N, Afifah R N, Zuaiini P A K and Rezapratama M S 2023 Analysis of ecotourism development as a mangrove conservation effort in Pasir Kadilangu and Jembatan Api-Api, Kulon Progo, Yogyakarta, Indonesia; *Indo Pac. J. Ocean Life* **7(2)**.

Bijsterveldt C Evan, Der Wal Dvan, Mancheño A G, Fivash G S, Helmi M and Bouma T J 2023 Can cheniers protect mangroves along eroding coastlines? – the effect of contrasting foreshore types on mangrove stability; *Ecol. Eng.* **187** 106,863.

Bimrah K, Dasgupta R, Hashimoto S, Saizen I and Dhyani S 2022 Ecosystem services of mangroves: A systematic review and synthesis of contemporary scientific literature; *Sustainability* **14(19)** 12,051.

Breiman L 2001 Random forests; *Mach. Learn.* **45** 5–32.

Breiman L, Friedman J, Olshen R and Stone C 1984 *Classification and regression trees* (Monterey, California: Wadsworth).

Budiyanto F, Alhomaidi E A, Mohammed A E, Ghandourah M A, Alorfi H S, Bawakid N O and Alarif W M 2022 Exploring the mangrove fruit: From the phytochemicals to functional food development and the current progress in the middle east; *Mar. Drugs* **20(5)** 303.

Carlson R R, Evans L J, Foo S A, Grady B W, Li J, Seeley M, Xu Y and Asner G P 2021 Synergistic benefits of conserving land–sea ecosystems; *Glob. Ecol. Conserv.* **28** e01,684.

Chicco D and Jurman G 2020 The advantages of the Matthews correlation coefficient (MCC) over F1 score and accuracy in binary classification evaluation; *BMC Genomics* **21(1)** 1–13.

Chicco D and Jurman G 2023a The Matthews correlation coefficient (MCC) should replace the ROC AUC as the standard metric for assessing binary classification; *BioData Min.* **16(1)** 4.

Chicco D and Jurman G 2023b A statistical comparison between Matthews correlation coefficient (MCC), prevalence threshold, and Fowlkes–Mallows index; *J. Biomed. Inform.* **144** 104,426.

Congalton R G 1991 A review of assessing the accuracy of classifications of remotely sensed data; *Remote Sens. Environ.* **37(1)** 35–46.

De Silva W and Amarasinghe M D 2023 Coastal protection function of mangrove ecosystems: A case study from Sri Lanka; *J. Coast. Conserv.* **27(6)** 59.

Demir S and Şahin E K 2022 Liquefaction prediction with robust machine learning algorithms (SVM, RF, and XGBoost) supported by genetic algorithm-based feature selection and parameter optimization from the perspective of data processing; *Environ. Earth Sci.* **81(18)** 459.

Estoque R C, Myint S W, Wang C, Ishtiaque A, Aung T T, Emerton L, Ooba M, Hijioka Y, Mon M S and Wang Z *et al.* 2018 Assessing environmental impacts and change in Myanmar's mangrove ecosystem service value due to deforestation (2000–2014); *Glob. Change Biol.* **24(11)** 5391–5410.

Faruque M J, Vekerdy Z, Hasan M Y, Islam K Z, Young B, Ahmed M T, Monir M U, Shovon S M, Kakon J F and Kundu P 2022 Monitoring of land use and land cover changes by using remote sensing and GIS techniques at

human-induced mangrove forests areas in Bangladesh; *Remote Sens. Appl.: Soc. Environ.* **25** 100,699.

Goswami A, Sharma D, Mathuku H, Gangadharan S M P, Yadav C S, Sahu S K, Pradhan M K, Singh J and Imran H 2022 Change detection in remote sensing image data comparing algebraic and machine learning methods; *Electronics* **11(3)** 431.

Islam K, Sarker S, Morsad G, Khan M F, Chowdhury A and Islam S L U 2024 Spatial–temporal changes of shoreline and vegetation: Impacts on mangrove cover along the Sundarbans area, Bangladesh; *J. Coast. Conserv.* **28(1)** 10.

Jardim A M d R F, Araújo Júnior G d N, Silva M V d, Santos A d, Silva J L B d, Pandorfi H, Oliveira-Júnior J F d, Teixeira A H d C, Teodoro P E and Lima J Lde *et al.* 2022 Using remote sensing to quantify the joint effects of climate and land use/land cover changes on the Caatinga biome of northeast Brazilian; *Remote Sens.* **14(8)** 1911.

Kadaverugu R, Dhyani S, Purohit V, Dasgupta R, Kumar P, Hashimoto S, Pujari P and Biniwale R 2022 Scenario-based quantification of land-use changes and its impacts on ecosystem services: A case of Bhitarkanika mangrove area, Odisha, India; *J. Coast. Conserv.* **26(4)** 30.

Kulkarni A D and Lowe B 2016 Random forest algorithm for land cover classification; *Int. J. Recent Innov. Trends Comput. Commun. (IJRITCC),* pp. 58–63.

Lawrence R L and Wright A 2001 Rule-based classification systems using classification and regression tree (CART) analysis; *Photogramm. Eng. Remote Sens.* **67(10)** 1137–1142.

Leo B, Friedman J H, Olshen R A and Stone C J 1984 Classification and regression trees; *Wadsworth International Group* **8** 452–456.

Li C, Wang F, Yang P, Wang Fc, Hu Yz, Zhao Yl, Tian Lz and Zhao Rb 2024 Mangrove wetlands distribution status identification, changing trend analyzation and carbon storage assessment of China; *China Geol.* **7(1)** 1–11.

Miraki M, Sohrabi H and Immitzer M 2023 Tree species mapping in mangrove ecosystems using UAV-RGB imagery and object-based image classification; *J. Indian Soc. Remote Sens.* **51(10)** 2095–2103.

Morales A and Villalobos F J 2023 Using machine learning for crop yield prediction in the past or the future; *Front. Plant Sci.* **14** 1128,388.

Nag S, Dutta R, Chakraborty S, Bagchi A, Mitra O, Samanta S and Pandey P 2024 Halotolerant and halophilic bacteria present in the mangrove ecosystem: Emerging bioengineering potentials; In: *Biotechnology of emerging microbes*; pp. 143–162.

Nagarajan P, Rajendran L, Pillai N D and Lakshmanan G 2022 Comparison of machine learning algorithms for mangrove species identification in Malad Creek, Mumbai, using Worldview-2 and Google Earth images; *J. Coast. Conserv.* **26(5)** 44.

Nguyen L T M, Hoang H T, Choi E and Park P S 2023 Distribution of mangroves with different aerial root morphologies at accretion and erosion sites in Ca Mau Province, Vietnam; *Estuar. Coast. Shelf Sci.* **287** 108,324.

Padhy S, Dash P and Bhattacharyya P 2022 Challenges, opportunities, and climate change adaptation strategies of mangrove-agriculture ecosystem in the Sundarbans, India: A review; *Wetl. Ecol. Manag.* **30(1)** 191–206.

Phan M H and Stive M J 2022 Managing mangroves and coastal land cover in the Mekong delta; *Ocean Coast. Manag.* **219** 106,013.

Priyono K D, Saifuddin A, Nugroho F S and Jumadi J 2022 Identification of mangrove changes in the Mahakam delta in 2007–2017 using ALOS/PALSAR and Landsat; *GEOMATE J.* **23(96)** 77–84.

Purwanto A D, Wikantika K, Deliar A and Darmawan S 2022 Decision tree and random forest classification algorithms for mangrove forest mapping in Sembilang National Park, Indonesia; *Remote Sens.* **15(1)** 16.

Quang N H, Quinn C H, Carrie R, Stringer L C, Hackney C R, Van Tan D *et al.* 2022 Comparisons of regression and machine learning methods for estimating mangrove aboveground biomass using multiple remote sensing data in the Red River estuaries of Vietnam; *Remote Sens. Appl.: Soc. Environ.* **26** 100,725.

Sameer M K and Hamid A M 2023 Remote sensing and GIS techniques in monitoring land useland cover change; *Int. J. Sustain. Constr. Eng. Technol.* **14(1)** 13–20.

Saravanan S, Abijith D, Reddy N M, Parthasarathy K, Janardhanam N, Sathiyamurthi S and Sivakumar V 2023 Flood susceptibility mapping using machine learning boosting algorithms techniques in Idukki district of Kerala, India; *Urban Clim.* **49** 101,503.

Shetty S 2019 Analysis of machine learning classifiers for LULC classification on Google Earth engine; Master's thesis; University of Twente.

Shindikar M 2006 Ecological studies on mangroves of Maharashtra coast; *An unpublished dissertafion submitted to University of Pune,* 189p.

Sikakwe G U 2023 Mineral exploration employing drones, contemporary geological satellite remote sensing and geographical information system (GIS) procedures: A review; *Remote Sens. Appl.: Soc. Environ.* 100988.

Silva W OBeys-da, Santi L and Guimarães J A *et al.* 2014 Mangroves: A threatened ecosystem under-utilized as a resource for scientific research; *J. Sustain. Dev.* **7(5)** 40–51.

Singh A 2019 Remote sensing and GIS applications for municipal waste management; *J. Environ. Manag.* **243** 22–29.

Singh V P, Singh R, Paul P K, Bisht D S and Gaur S 2024 Remote sensing and geographic information systems driven data analysis; In: *Hydrological Processes Modelling and Data Analysis: A Primer*, pp. 73–103.

Soni P K, Rajpal N, Mehta R and Mishra V K 2022 Urban land cover and land use classification using multispectral Sentinal-2 imagery; *Multimed. Tools Appl.* **81(26)** 36,853–36,867.

Sulochanan B, Ratheesh L, Veena S, Padua S, Prema D, Rohit P, Kaladharan P and Kripa V 2022 Water and sediment quality parameters of the restored mangrove ecosystem of Gurupura river and natural mangrove ecosystem of Shambhavi river in Dakshina Kannada, India; *Mar. Pollut. Bull.* **176** 113,450.

Toosi N B, Soffianian A R, Fakheran S and Waser L T 2022 Mapping disturbance in mangrove ecosystems: Incorporating landscape metrics and PCA-based spatial analysis; *Ecol. Indic.* **136** 108,718.

Trialfhianty T I, Muharram F W, Quinn C H, Beger M *et al.* 2022 Spatial multicriteria analysis to capture socio-

economic factors in mangrove conservation; *Mar. Policy* **141** 105,094.

Wang J, Bretz M, Dewan M A A and Delavar M A 2022 Machine learning in modelling land-use and land cover-change (LULCC): Current status, challenges and prospects; *Sci. Total Environ.* **822** 153,559.

Wang Z, Wang E and Zhu Y 2020 Image segmentation evaluation: A survey of methods; *Artif. Intell. Rev.* **53(8)** 5637–5674.

Wieckowska B, Kubiak K B, Jóźwiak P, Moryson W and Stawińska-Witoszyńska B 2022 Cohen's Kappa coefficient as a measure to assess classification improvement following the addition of a new marker to a regression model; *Int. J. Environ. Res. Public Health* **19(16)** 10,213.

Wulder M A, Roy D P, Radeloff V C, Loveland T R, Anderson M C, Johnson D M, Healey S, Zhu Z, Scambos T A and Pahlevan N *et al.* 2022 Fifty years of Landsat science and impacts; *Remote Sens. Environ.* **280** 113,195.

Xia Y, Watts J D, Machmuller M B and Sanderman J 2022 Machine learning based estimation of field-scale daily, high resolution, multi-depth soil moisture for the western and midwestern United States; *PeerJ* **10** e14,275.

Xu H and Deng W 2022 Rationality analysis of MRSEI and its difference with RSEI; *J. Remote Sens. Technol.* **37(1)** 1–7.

Yang G, Huang K, Sun W, Meng X, Mao D and Ge Y 2022 Enhanced mangrove vegetation index based on hyperspectral images for mapping mangrove; *ISPRS J. Photogramm. Remote Sens.* **189** 236–254.

Yao J and Shepperd M 2020 Assessing software defection prediction performance: Why using the Matthews correlation coefficient matters; *Proceedings of the Evaluation and Assessment in Software Engineering*, pp. 120–129.

Zhao C, Jia M, Wang Z, Mao D and Wang Y 2023 Identifying mangroves through knowledge extracted from trained random forest models: An interpretable mangrove mapping approach (IMMA); *ISPRS J. Photogramm. Remote Sens.* **201** 209–225.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.

Corresponding editor: Saumitra Mukherjee